

JOAQUIN AUZENNE

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EDUCATION

University of Texas at Austin

Austin, TX

B.S. Mathematics, B.S. Computational Biology, Certificate in Scientific Computation and Data Sciences

May 2025

- College of Natural Sciences Polymathic Scholars Honors
- Thesis: *DNA Metabarcoding of Soil Samples from the Magdalena River Valley*

University of North Texas

Denton, TX

Pre-major studies in Biology and Chemistry (transferred to UT Austin)

Aug. 2019 – May 2021

RESEARCH EXPERIENCE

Bioinformatic Assistant

Austin, TX

DiFiore Lab, University of Texas at Austin

Dec. 2024 – Present

- Developed a mammalian metabarcoding pipeline for pre-processing and consensus sequence clustering of Nanopore MinION-sequenced environmental DNA (83 soil samples, 6M reads) in Shell script for taxonomic assignment (BLAST).
- Integrated multiple public domain databases (NCBI, IUCN) using the GBIF Metabarcoding Data Toolkit API for comprehensive species identification and interactive species-range visualization (ArcGIS).
- Established a curated DNA reference library of mammals identified along the Magdalena River in Colombia, focusing on rRNA genetic markers (12S, 16S).
- Delivered consensus sequences ready for longitudinal population and dispersal analyses using MinION MKICN Nanopore sequencing, MultiQC, NGSspeciesID read clustering, and megaBLAST taxonomic assignment.

Undergraduate Researcher & Teaching Mentor

Austin, TX

FRI EvoDevOmics, University of Texas at Austin

Jan. 2022 – Sept. 2023

- Designed high-throughput transcriptomic pipelines (preprocessing, quality control, alignment) in R on BRCF HPC clusters to reveal the basis of embryonic diversification in Dendrobatid frogs via differential gene expression analyses (DESeq2, WGCNA, PCA, nearest-neighbors clustering).
- Employed Python tool IdMiner for automated literature searches through the PaperBLAST database (PubMed, NCBI) to functionally annotate 40+ differentially expressed genes.
- Served as Teaching Assistant for a computational biology course of 20+ students, covering next-generation sequencing, omics data analysis, FAIR digital resource stewardship, and career readiness under Rebecca Young, PhD.
- Presented weekly lessons on statistical methods (MANOVA, PCA, WGCNA) to student cohorts, translating complex bioinformatic concepts for undergraduate audiences.

Undergraduate Research Assistant

Denton, TX

Henard Lab, University of North Texas BioDiscovery Institute

Feb. 2020 – May 2021

- Awarded \$500 competitive grant to design a shikimate biosynthesis pathway in plasmid-modified *E. coli* (Gibson assembly) as a “green chemistry” alternative for nylon precursor production in a scalable methane-fed bioreactor.
- Maintained *E. coli* and methanotrophic cultures; evaluated methane production within continuous-stirred tank reactors; performed qRT-PCR confirmation of engineered shikimate-pathway constructs.
- Conducted daily hood upkeep, sterility checks, liquid chromatography analysis, and detailed laboratory record-keeping.
- Adapted “*Muconic acid production from methane using rationally-engineered methanotrophic biocatalysts*” (Henard et al.) for *Frontiers for Young Minds*, translating advanced metabolic engineering concepts for a younger audience.

Reptile Care & Research Assistant

Denton, TX

Crossley Lab, University of North Texas Integrative Biology

Oct. 2019 – Jun. 2020

- Oversaw daily feeding, tank maintenance, and husbandry for 30+ tanks of large laboratory reptiles (*Alligator mississippiensis*, *Chelydra serpentina*); provided on-call emergency support including animal restraint, transport, and surgical assistance.
- Conducted cardiorespiratory physiological study of alligators and snapping turtles birthed under normoxic and hypoxic conditions via controlled treadmill exercise and blood gas centrifugation.

PROFESSIONAL EXPERIENCE

Laboratory Operator III

Emerald Cloud Lab

Austin, TX

Jan. 2026 – Present

- Execute precision experimental protocols across automated liquid handling, robotic instrumentation, and high-throughput assay platforms in a GMP/GLP-compliant cloud laboratory environment serving multiple simultaneous client projects.
- Operate genomics-adjacent instrumentation including qPCR thermocyclers, acoustic liquid handlers for nanoliter-scale nucleic acid transfers, capillary gel electrophoresis systems for fragment analysis and library QC, flow cytometers, and Sanger sequencing platforms — building wet-lab fluency directly upstream of the NGS library preparation and sequencing workflows central to RNA-seq and eDNA metabarcoding research.
- Monitor and interpret real-time QC metrics across automated workflows, flagging anomalous results and coordinating with internal teams to ensure instrument calibration and data quality.

Data Analysis Intern

Tanel Health (Heman Sweatt Center for Black Males, UT Austin)

Dakar, Senegal

May 2023 – Aug. 2023

- Implemented gradient boosting and regression (XGBoost, SARIMA) on pharmaceutical sales data to improve back-end inventory forecasting on AWS, with projected waste reduction of up to 20% across 200+ in-network enterprises.
- Created technical reports justifying predictive model selections to shareholders, including associated performance metrics and visualizations (Python: matplotlib, Plotly).
- Digitized prescriptions and integrated a two-factor QR-based prescription authentication system into a 10,000+ customer database, improving compliance and fraud prevention.
- Collaborated with an interdisciplinary team of healthcare providers and software developers in a cross-cultural environment, adhering to strict data privacy standards.

Consultant Machinist

Paradigm Robotics

Austin, TX

Jan. 2024 – Aug. 2024

- Fabricated and assembled mechanical components for fire-resistant robotic systems (FireBot), translating CAD designs into G-code and operating mill/lathe equipment; collaborated with engineers on design iterations and equipment modifications.

CNC Machinist & Programmer

Carr Lane Manufacturing

Austin, TX

Aug. 2021 – Nov. 2022

- Trained new CNC machinists, performed GD&T quality inspections, and maintained 5S standards while machining on HAAS/DMG Mori mills and lathes; programmed overnight production runs increasing weekly machining output by 24+ hours.
- Led evening shift operations, maintaining detailed maintenance logs for tool performance, inventory tracking, and equipment status; documented procedures in SOPs for consistent quality and safety compliance.

SKILLS

Bioinformatics: Illumina (NextSeq, NovaSeq), ONT (MinION, MKICN), QC (Cutadapt, FastQC/MultiQC, NanoPlot), Alignment/Clustering (STAR/Salmon, Kallisto, NGSspeciesID, BWA-MEM2), RNA-Seq (WGCNA, DESeq2), Variant Calling, AlphaFold2

Programming: Python (pandas, scikit-learn, PyTorch, TensorFlow, Biopython), R (tidyverse, ggplot2, Shiny, MCMCglmm, nlme), SQL, Shell Scripting (Bash), MATLAB, High Performance Computing (SLURM), Java, HTML/CSS, C++

Data Analysis: Statistical Modeling (ANCOVA, MLE, PCA, XGBoost, SARIMA), Visualization (ggplot2, matplotlib, Plotly), NLP, GenAI (LLM RAG), Cloud Computing (AWS), FAIR Data Management

Wet Lab: DNA/RNA Extraction, PCR/qPCR/ddPCR, SDS-PAGE, Anaerobic & Microaerophilic Culture (Coy chamber), CRISPR/Cas-Mediated iPSC, Plasmid Design (Gibson Assembly), Light Microscopy, Blood Gas Analysis, GMP/GLP Compliance

Tools & Platforms: LIMS (Benchling), Docker (Bioconductor), Linux, Git/GitHub, VSCode, L^AT_EX, ArcGIS, MS Office, Adobe Acrobat

Languages: English (Native), French (Fluent), Spanish (Conversational), Japanese (Novice)

PROJECTS

Mammalian Metabarcoding & Species-Range Visualization Platform

DiFiore Lab, University of Texas at Austin

Austin, TX

Dec. 2024 – Present

- Developed an interactive web application presenting Nanopore eDNA genomic analyses as user-friendly interfaces for both scientific and public audiences, integrating sequencing results with geospatial species-range data for the Magdalena River Valley.

COVID-19 Spatiotemporal Network Dynamics

Department of Integrative Biology, University of Texas at Austin

Austin, TX

Oct. 2024 – Dec. 2024

- Built spatial network time-series models of U.S. COVID-19 spread (32M cases, 04/2020–04/2021) with Shiny interactive maps, dashboards, and statistics to identify transmission hubs from weighted network centrality to inform public-health intervention strategies.

IdMiner — Bioinformatic Literature Search Tool

FRI EvoDevOmics, University of Texas at Austin

Austin, TX

2022

- Co-developed a Python-based academic literature search application querying PubMed and NCBI repositories for gene-associated publications to aid researchers in functional annotation of gene lists at scale.

Methanotroph Metabolic Engineering

Henard Lab, University of North Texas BioDiscovery Institute

Denton, TX

Feb. 2020 – May 2021

- Awarded \$500 to design a shikimate biosynthesis pathway for plasmid-modified *E. coli* (Gibson assembly) for “green chemistry” alternative nylon production in a scalable methane-fed bioreactor.
- Adapted “*Muconic acid production from methane using rationally-engineered methanotrophic biocatalysts*” (Henard et al.) for *Frontiers for Young Minds*, translating advanced metabolic engineering concepts for a younger audience.

LEADERSHIP, MENTORSHIP & SERVICE

EvoDevOmics Peer Mentor (UT Austin FRI, 2022–2023): Mentored 20+ undergraduates in computational biology, omics data analysis, NGS methods, and FAIR data stewardship; fostered independent research skills and career readiness.

Sweatt in Senegal Program (UT Austin, Summer 2023): Selected for inaugural cohort; led cross-cultural data science collaboration in Dakar, Senegal, contributing to healthcare informatics projects in a resource-limited international setting.

Science Communication: Adapted a peer-reviewed metabolic engineering research for *Frontiers for Young Minds*.

CERTIFICATIONS & TRAINING

MSSC Certified Production Technician

Lean Six Sigma White Belt

OSHA-10 General Industry Safety Certification

PUBLICATIONS & PRESENTATIONS

COVID-19 Spatiotemporal Network Dynamics — Course presentation, Department of Integrative Biology, UT Austin, December 2024.

REFERENCES

Available upon request. Faculty and supervisor references available from: Dr. Anthony Di Fiore (UT Austin), Dr. Rebecca Young (UT Austin), Dr. Calvin Henard (UNT BioDiscovery Institute).